

Peer Review Report

Title: Multi-sensor dataset of a tendon-driven continuum robot in dynamic motion.

Recommendation: Major revision.

Summary

The work presents a dataset for a single-segment soft robot, built from commercially available components. The dataset contains readings from six sensor modalities in quasi-static and dynamic motions as well as contact situation. Cross-validation was undertaken for consistency assessment. Overall, the work addresses a general lack of comparable, transferable, standardized and available experimental data in the field of soft robotics. The scientific contribution and relevance of the work is well motivated and aims at filling an existing research gap in the field of soft robotics. However, the paper and the dataset show major weaknesses and inconsistencies, which should be revised before the paper can be published. Details below.

Major comments (paper)

- The introduction to the work is thoroughly written and also effectively compares the state of the art in soft robotics well with other data-intensive research fields such as autonomous driving and robotics in general. However, the introduction lacks comparisons to datasets in the same field, which should be added/referred to, e.g.,
 - H. M. Hansen, B. K. Sæbø, K. Y. Pettersen, J. T. Gravdahl, M. Di Castro, "Data-Driven Dynamic Modeling of a Tendon-Actuated Continuum Robot," arXiv:2605.18720, 2026, DOI: <https://doi.org/10.48550/arXiv.2605.18720>
 - C. Muhmann, R. M. Grassmann, M. Bartholdt, J. Burgner-Kahrs, "Toward Dynamic Control of Tendon-Driven Continuum Robots using Clarke Transform," IROS 2025, DOI: 10.1109/IROS60139.2025.11247518.
 - G. Zhang, Z. Chen, L. Wang, "Virtual-Work Based Shape-Force Sensing for Continuum Instruments with Tension-Feedback Actuation," arXiv:2501.05418, 2025.
 - Y. Liu, K. Luo, Q. Tian, H. Hu, "An efficient data-driven framework of hybrid dynamics for real-time modeling and control of continuum robots," Mechanism and Machine Theory 220, 106361, 2026. DOI: 10.1016/j.mechmachtheory.2026.106361.
 - E. Wang, Z. Deng, C. Pan, B. He, J. Zhang, "Estimating Continuum Robot Shape under External Loading using Spatiotemporal Neural Networks," IROS 2025. DOI: 10.1109/IROS60139.2025.11245975.
- While the experimental setup is described in detail, the figures (especially Fig. 2) should be much clearer:
 - Inconsistent writing (lowercase / uppercase). Example: Contact probe vs. Reflective Marker.
 - Coordinate frames are barely visible, especially when printed on paper in greyscale. Increase contrast.
 - Dashed lines and circles are not aligned well. Correct alignment.
- Multiple qualitative terms such as "rigidly connected" and "geometrically stable" are used. Such vague terms are too unprecise in an experimental paper about soft robotics, which heavily relies on a sufficiently rigid mounting structure. The authors should provide

quantitative measurements such as static directional stiffness values or structural eigenfrequencies to quantify that the mounting structure is sufficiently stiff for the experiments.

- The contact-force validation remains largely qualitative. Terms such as “close agreement” and “good agreement” should be supported by quantitative metrics such as RMSE, mean bias, maximum absolute error, and preferably correlation values.
- Inconsistency in Tab 1: The table does not only contain geometrical parameters. Correct the description of the table.
- Inconsistency in Tab 1: The physical unit of weight is newtons (N), not grams (g). Refer to mass, not weight, when using grams as the physical unit. Correct units.
- Inconsistency with kinematic definitions: g_{fix} is introduced as a rotation matrix (orientation), but g_{wand} is introduced as a pose. In robotics, a pose is position and orientation.
- The paper currently lacks a clear, dedicated conclusion and summary section. The authors should add a concluding section that summarizes the impact of the dataset and outlines potential future use cases.
- The paper currently lacks an application to a baseline model for at least one of the use cases that are mentioned in the introduction. Therefore, the paper does not demonstrate by itself that the dataset is useful for any of these applications. Neither does it demonstrate that the authors have successfully used their dataset. This could have helped with the inconsistencies of the dataset (see further below).
- The GVS model used for figure 6 and table 7 is not included in the released code. Since this model forms part of the technical validation, the implementation and all parameters required to reproduce the reported results should be released.

Minor comments (paper)

- The word “minimize” should only be used when referring specifically to the results of an optimization routine. Consider using “reduce” or “eliminate” to avoid confusion. This occurs multiple times in the paper and should be corrected.
- Sec 2.2: 3D printed pulleys appear to be an unnecessarily low-cost option for routing cables. Have the authors checked for roundness (part may have a seam from printing) and eccentricity after pressing the bearing in? It remains unclear, why 3D printed pulleys were used at all, as machined parts are widely available standard components.
- While mathematically identical for orthogonal matrices, it is rather common in robotics to use the transpose of a rotation matrix instead of its inverse (see inversion of g_{base} on page 16). The authors may want to use the transpose for stylistic consistency.
- Some parts of the paper read more like a technical documentation rather than a scientific paper. Overall, the writing could be improved before publication so that the text reads more fluently.
- Data availability and code availability refer to the same internet source, so it could be included in a single section rather than two short sections.
- Table 3 states that the FBG shape contains 502 points but uses the index range 0–502, which would correspond to 503 points.

Major comments (dataset)

- The documented column order of `mocap_frames.csv` appears to be incorrect. Table 3 states [x, y, z, roll, pitch, yaw], whereas both the released processing code (`data_optitrack.m`, lines 143–149; `process_data.m`, lines 1049–1066) and inspection of the released CSV indicate

[roll, pitch, yaw, x, y, z]. This can directly lead to incorrect interpretation of the dataset and should be corrected before publication.

- A Python logic error is present in `read4MotorCircle.py`: if `motion_pattern == "circle" or "Lissajous"`: This condition is always true because the non-empty string literal "Lissajous" evaluates to true in a Boolean context in Python. As a consequence, the angular tolerance is apparently set to 12° for all motion types rather than the intended 0.5° default. The authors should clarify whether this code version was used for the published experiments and assess its impact on the recorded trajectories.
- The filtering procedure may be seriously affected by irregular timestamps. `process_data.m` determines the sampling frequency using $F_s = 1/\text{median}(\text{diff}(t))$ and applies a standard Butterworth filter before uniform resampling. This assumes approximately equidistant samples. The inspected ATI raw recording exhibits strongly burst-like host timestamps despite an average data rate of approximately 200 Hz, meaning that the median timestamp interval may yield a grossly incorrect filter sampling frequency. The authors should verify whether the released processed signals were affected and, if necessary, regenerate them using a regular time grid before filtering.
- Contact-data processing is inconsistent between the manuscript and the released code. The manuscript specifies a 15 Hz low-pass filter, whereas `compare_ft_sensors.m` uses 30 Hz. Processed files are described as header-less, while the contact script uses `writetable`. File names also differ (`tendon_tensions.csv` VS. `cable_tensions.csv`, `contact_wrench.csv` VS. `wrench_wand.csv`), and no clear generator for the documented `wand_pose.csv` was identified.
- The main acquisition script does not reproduce the contact subset. `runAllDataCollection.bat` starts FBG, ATI, OptiTrack, four Mark-10 processes, motors, and an undocumented gyroscope, but does not start the Resense/HEX12 acquisition process required for the contact experiments.
- The released code is not yet a fully reproducible standalone processing pipeline. Important parameters and paths are hard-coded in `process_data.m`, no complete top-level README or dependency file is provided, and several required software/hardware dependencies are not documented. A clean, parameterized Raw → Processed workflow should be provided. The manuscript and the released processing code describe different OptiTrack calibration procedures. Section 2.6.3 states that the per-disk marker offsets are determined once from `references/straight_config`, whereas `process_data.m` (lines 141–195) recalculates these offsets independently from the first 3 s of every recording. The authors should clarify which procedure was actually used to generate the released processed data.
- The description of the circle and Lissajous trajectories does not fully match the acquisition code. The manuscript describes continuous sinusoidal commands, whereas `read4MotorCircle.py` implements the circle using discrete target configurations and the Lissajous trajectory using 16 sampled waypoints that are sequentially executed by position control. The actual actuation protocol should be described accurately.
- The quasi-static acquisition protocol cannot be reproduced from the released acquisition code. The manuscript describes small incremental sweeps through near-equilibrium configurations, while the available code mainly contains two extreme target configurations and no explicit equilibrium criterion. Either the actual acquisition script should be released or the Methods section should be revised accordingly.
- The temporal synchronization discussion appears too strong. Although all sensor processes use the same host clock, their timestamps correspond to different events such as serial-response completion, DAQ block reads, sequential CAN queries, network

reception, or processed FBG-frame reception. A common OS clock therefore does not necessarily imply a common physical measurement time.

- The manuscript states that cross-correlation was omitted for tendon tension and base force/torque signals, while the released synchronization code nevertheless computes motor-tension and motor-ATI correlations. The authors should clarify whether these were exploratory analyses only and align the released validation code with the description in the paper.
- The FBG-OptiTrack comparison should be described more carefully as a cross-modal consistency check after frame registration. Since OptiTrack data are used to align the FBG frame before the two modalities are compared, this is not a completely independent absolute validation.

Minor comments (dataset)

- The OptiTrack validity flags (`disk_k_is_valid`) are loaded in `data_optitrack.m` but apparently not used during processing, and the validity information is not retained in the processed data. The handling of occluded/invalid frames should be documented and preferably retained as a validity mask.
- The FBG delay value is missing from the Methods text (“a fixed offset of is subtracted”), while the code uses approximately 13.5 ms. The numerical value and sign convention should be stated explicitly.
- Delay estimation and delay correction are mixed in `process_data.m`: the FBG timestamps are corrected before the synchronization check is executed. The lag estimation should preferably be performed on the uncorrected signals and separated from the subsequent correction step.
- The manuscript refers to `std::chrono::high_resolution_clock` for FBG timestamps, whereas the released C++ code stores timestamps using `std::chrono::system_clock`.
- All four sequentially queried motor states are stored using a single timestamp. Either individual timestamps should be used or the total four-motor readout latency should be quantified.
- The ATI timestamp is assigned after reading and averaging a five-sample DAQ block. The timestamp therefore represents approximately the end rather than the center of the measurement block.
- The manuscript specifies a 2 s pre-motion period, while `read4MotorCircle.py` uses 3 s. This should be made consistent.
- The common 100 Hz resampling interval is based primarily on the motor recording duration rather than the common overlap of all sensor streams. The authors should verify that this does not introduce undefined or NaN edge samples.
- Processed signals are generated using zero-phase `filtfilt`, which is acausal. This is appropriate for offline analysis but should be explicitly stated in the Usage Notes because the dataset is also proposed for online/state-estimation applications.
- The reported Mark-10 noise values should be checked. `analyze_reference_noise.m` appears to calculate the standard deviation from the raw pulley reaction force, while normal processing divides these values by two to obtain tendon tension.
- The “Euclidean distance” used for contact-position validation is calculated only in the xy plane in `compare_ft_sensors.m`. It should either be referred to as planar distance or calculated in full 3-D.

- The FBG files referred to as “raw” already contain proprietary Shape CORE reconstructed curvature, bending angle, and 3-D shape rather than raw optical FBG measurements. The terminology should be clarified.
- The statement that processed FBG quantities are retained “without further modification” is inaccurate because units, coordinate frames, timestamps, filtering, and sampling are modified during post-processing.
- Several additional code/documentation inconsistencies should be cleaned up, including incorrect torque units in plot labels, obsolete function calls, inconsistent comments on FBG column layout, and incomplete MATLAB/Python dependency documentation.

Summary

Overall, the paper addresses an existing research gap in the field of soft robotics. The primary motivation and the targeted scientific contribution are clear and valid. However, the paper and the dataset need more thoroughness and carefulness to reliably deliver the intended contribution and allowing others to use the dataset. A more precise use of terms and definitions is required in the manuscript. Also, several inconsistencies between the manuscript, the released acquisition/processing code, and the documented data structure need to be resolved. These include the interpretation of processed motion-capture data, calibration procedures, trajectory generation, temporal synchronization, filtering, and the reproducibility of the released processing pipeline. The dataset appears promising, but these issues should be corrected to ensure reliable third-party reuse.

Recommendation

A major revision is necessary before the paper and the dataset can be published.